3 State Discrimination Using Gaussian States

3.1 Introduction

With the basic formalism and notation for continuous variable states established, we now address the problem of discriminating between such states, focusing particularly on Gaussian states.

A continuous variable state is classified as Gaussian if its Wigner function – or equivalently, its characteristic function – takes a Gaussian form. Mathematically, this is expressed as:

$$W[\hat{\rho}](\vec{R}) = \frac{1}{(2\pi)^n\sqrt{\det\sigma}} \exp\left(-\frac{1}{2}(\vec{R}-\bar{R})^T\sigma^{-1}(\vec{R}-\bar{R})\right) \tag{3.1}$$

where,

- $W[\hat{\rho}](\vec{R})$: The Wigner function tells us the 'probability-like' value at point $\vec{R}$ in phase space.
- $\vec{R}$: This is a vector containing the positions and momenta of all n modes. For example, for one mode, $\vec{R} = (q, p)$.
- $\bar{R}$: The average (mean) value of $\vec{R}$. This is the center of the bell curve.
- σ: The covariance matrix. It tells us how spread out the state is in different directions.
- $\det\sigma$: The determinant of the covariance matrix. It measures the overall volume of the distribution.
- n: The number of modes (like the number of particles or light beams).

Alternatively,

$$\chi[\hat{\rho}](\vec{\Lambda}) = \exp\left(-\frac{1}{2}\vec{\Lambda}^T\sigma\vec{\Lambda} + i\vec{\Lambda}^T\bar{R}\right) \tag{3.2}$$

where,

- $\chi[\hat{\rho}](\vec{\Lambda})$: The characteristic function. It is like the 'moment-generating' function in statistics.
- $\vec{\Lambda}$: A vector in phase space that is conjugate to $\vec{R}$ (like frequency is conjugate to time).
- σ: The same covariance matrix from Equation (3.1).

- $\bar{R}$: The same mean vector from Equation (3.1).

Thus, a Gaussian state $\hat{\rho}$ is completely characterized by its covariance matrix σ and its mean vector $\bar{R}$. The covariance matrix gains further physical significance through Williamson's Theorem, which decomposes it into a symplectic matrix and a diagonal matrix of symplectic eigenvalues. Importantly, all entanglement properties of a Gaussian state are contained within σ. This is because local operations do not affect entanglement, so the mean vector $\bar{R}$ can be arbitrarily adjusted through phase space translations without altering the entanglement of $\hat{\rho}$. Conversely, generating entanglement necessarily requires a nonlocal operation on $\hat{\rho}$.

3.2 Normal form decomposition

A useful theorem for expressing the covariance matrix of a Gaussian state is Williamson's Theorem, stated as follows:

Theorem 3.1 (Williamson's Theorem)**.** *Given a real $2n \times 2n$ symmetric and positive matrix σ, there exists a real and symplectic $2n \times 2n$ matrix S and a real $n \times n$ diagonal and positive matrix W such that*

$$\sigma = S^T W S \tag{3.3}$$

where

$$W = \bigoplus_{k=1}^{n} d_k \mathbb{I}_2 \tag{3.4}$$

and

$$\mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{3.5}$$

Proof. The proof proceeds in several steps. We will construct the symplectic matrix S that diagonalizes σ in the symplectic sense.

Since σ is real, symmetric, and positive-definite, it can be diagonalized by an orthogonal matrix $O \in O(2n)$:

$$\sigma = O^T D O, \tag{3.6}$$

where $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{2n})$ with $\lambda_i > 0$. However, this is an orthogonal diagonalization, not a symplectic one. Our goal is to find a symplectic matrix S that brings σ to the block-diagonal form W.

Define the matrix

$$M = i\Omega\sigma, \tag{3.7}$$

where Ω is the symplectic form:

$$\Omega = \bigoplus_{k=1}^{n} \omega, \qquad \omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{3.8}$$

Consider the matrix $M = i\Omega\sigma$. Its transpose is:

$$M^T = (i\Omega\sigma)^T = i\sigma^T\Omega^T = i\sigma(-\Omega) = -i\sigma\Omega. \tag{3.9}$$

Now note that:

$$\Omega^{-1}M\Omega = \Omega^{-1}(i\Omega\sigma)\Omega = i\sigma\Omega. \tag{3.10}$$

Since $\sigma\Omega$ is similar to $-\Omega\sigma$ (because $\Omega^T = -\Omega$), the eigenvalues of M come in $\pm$ pairs. With the factor i, the eigenvalues are of the form $\pm id_k$ where $d_k > 0$.

Let the eigenvalues of M be $\pm id_1, \pm id_2, \ldots, \pm id_n$ with $d_k > 0$. For each eigenvalue id_k, there exists a complex eigenvector u_k such that:

$$i\Omega\sigma u_k = id_k u_k \quad \Rightarrow \quad \Omega\sigma u_k = d_k u_k. \tag{3.11}$$

Similarly, for eigenvalue $-id_k$, we have an eigenvector v_k satisfying $\Omega\sigma v_k = -d_k v_k$.

Using the fact that σ is symmetric and Ω is skew-symmetric, one can show that v_k is proportional to the complex conjugate of u_k. We choose them such that:

$$v_k = i\sigma u_k. \tag{3.12}$$

This choice ensures the correct normalization.

We normalize the eigenvectors such that:

$$u_k^\dagger \sigma u_k = 1. \tag{3.13}$$

Then, using properties of Ω and σ, we also get:

$$u_k^\dagger \Omega u_k = i. \tag{3.14}$$

Define the real vectors x_k and y_k for $k = 1, \ldots, n$ as:

$$x_k = \sqrt{2}\,\mathrm{Re}(u_k), \qquad y_k = \sqrt{2}\,\mathrm{Im}(u_k). \tag{3.15}$$

One can verify that these vectors satisfy:

$$x_k^T \sigma x_k = y_k^T \sigma y_k = d_k, \tag{3.16}$$
$$x_k^T \sigma y_k = 0, \tag{3.17}$$

and

$$x_k^T \Omega x_k = y_k^T \Omega y_k = 0, \tag{3.18}$$
$$x_k^T \Omega y_k = 1. \tag{3.19}$$

Let S be the $2n \times 2n$ matrix whose columns are, in order:

$$S = \begin{pmatrix} x_1 & y_1 & x_2 & y_2 & \cdots & x_n & y_n \end{pmatrix}. \tag{3.20}$$

From the properties above:

- Symplectic condition: $S^T \Omega S = \Omega$ (since $x_k^T \Omega y_k = 1$ and all other pairings are zero).
- Diagonalization of σ: $S^T \sigma S = \bigoplus_{k=1}^n d_k \mathbb{I}_2 = W$.

Thus, we have constructed a symplectic matrix S such that:

$$S^T \sigma S = W \quad \Rightarrow \quad \sigma = (S^{-1})^T W S^{-1}. \tag{3.21}$$

Since S is symplectic, S^{-1} is also symplectic. Renaming S^{-1} as S (or absorbing the inverse into the definition), we obtain:

$$\sigma = S^T W S \tag{3.22}$$

with $S \in Sp(2n, \mathbb{R})$ and W diagonal as required. This completes the proof. □

The elements d_k of the matrix W are called the symplectic eigenvalues and can be calculated from the spectrum of $i\Omega\sigma$, while the matrix S is the symplectic matrix that performs the symplectic decomposition. Finding the symplectic eigenvalues of a covariance matrix is usually straightforward, but determining the symplectic matrix that performs the decomposition is often difficult. Performing the symplectic decomposition of Gaussian states is a key part of the process of determining the distinguishability of those states. For example, it turns out that in order to calculate the quantum Chernoff bound for Gaussian states – an asymptotic distinguishability measure for states that will be used extensively – we must find the symplectic decomposition of the states involved.

The phenomenological statement implied by equation (3.24) is that every Gaussian state $\hat{\rho}_G$ can be obtained from a thermal state $\hat{\rho}_{\text{th}}$ (described by a diagonal covariance

matrix as in equation (2.68)) by performing the unitary transformation U_S associated with the symplectic matrix S that performs the symplectic decomposition:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{3.23}$$

where $\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \hat{\rho}_{\text{th},2} \otimes \cdots \otimes \hat{\rho}_{\text{th},n}$ is a product of thermal states $\hat{\rho}_{\text{th},k}$ for $k = 1, \ldots, n$. Now, we derive the above equation.

From Williamson's Theorem, we have:

$$\sigma = S^T W S \tag{3.24}$$

where:

- σ is the covariance matrix of an arbitrary Gaussian state $\hat{\rho}_G$
- S is a symplectic matrix ($S \in Sp(2n, \mathbb{R})$)
- W is a diagonal matrix of the form:

$$W = \bigoplus_{k=1}^{n} d_k \mathbb{I}_2, \qquad \mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{3.25}$$

For a single-mode thermal state $\hat{\rho}_{\text{th},k}$ with average photon number N_k, the covariance matrix is (from equation 2.67 and 2.68):

$$\sigma_{\text{th},k} = \frac{1}{4} \begin{pmatrix} 2N_k + 1 & 0 \\ 0 & 2N_k + 1 \end{pmatrix} = \left(N_k + \frac{1}{2} \right) \mathbb{I}_2 \tag{3.26}$$

Comparing this with the block form of W, we identify:

$$d_k = N_k + \frac{1}{2} \tag{3.27}$$

Thus, the matrix W can be written as:

$$W = \bigoplus_{k=1}^{n} \left(N_k + \frac{1}{2} \right) \mathbb{I}_2 \tag{3.28}$$

This is exactly the covariance matrix of a **product of thermal states**:

$$\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \hat{\rho}_{\text{th},2} \otimes \cdots \otimes \hat{\rho}_{\text{th},n} \tag{3.29}$$

whose covariance matrix is:

$$\sigma_{\text{th}} = W \tag{3.30}$$

Now, we want to find a unitary transformation U_S that maps the thermal state $\hat{\rho}_{\text{th}}$ to the target Gaussian state $\hat{\rho}_G$.

In phase space, a unitary transformation generated by a quadratic Hamiltonian corresponds to a symplectic transformation on the covariance matrix:

$$\sigma \to S\sigma S^T \tag{3.31}$$

More precisely, if $\hat{\rho}' = U_S \hat{\rho} U_S^\dagger$, then the covariance matrix transforms as:

$$\sigma' = S\sigma S^T \tag{3.32}$$

where S is the symplectic matrix associated with the unitary U_S.

Let us apply the symplectic transformation S to the thermal state $\hat{\rho}_{\text{th}}$:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{3.33}$$

The covariance matrix of $\hat{\rho}_G$ becomes:

$$\sigma_G = S\sigma_{\text{th}} S^T \tag{3.34}$$

But $\sigma_{\text{th}} = W$, so:

$$\sigma_G = SWS^T \tag{3.35}$$

Williamson's Theorem states:

$$\sigma = S^T W S \tag{3.36}$$

Notice the difference: Williamson's theorem has $S^T W S$, while we have SWS^T. However, since S is symplectic, its inverse is given by:

$$S^{-1} = \Omega S^T \Omega^{-1} \tag{3.37}$$

More simply, if S is symplectic, then S^T is also symplectic. Therefore, we can rename S^T as a new symplectic matrix S'. This means:

$$SWS^T = (S^T)^T W (S^T) = S'^T W S' \tag{3.38}$$

Thus, both forms are equivalent. Williamson's theorem guarantees the existence of some symplectic matrix such that $\sigma = S^T W S$. Therefore, there exists a unitary transformation U_S such that:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{3.39}$$

Thus, we arrive at the equation:

$$\hat{\rho}_G = U_S \hat{\rho}_{\text{th}} U_S^\dagger \tag{3.40}$$

where:

- $\hat{\rho}_G$ is any Gaussian state
- $\hat{\rho}_{\text{th}} = \hat{\rho}_{\text{th},1} \otimes \cdots \otimes \hat{\rho}_{\text{th},n}$ is a product of thermal states
- U_S is the unitary transformation associated with the symplectic matrix S

3.3 Distinguisability Of States

Although, the problem of distinguisability is defined for any number of states, let us start with the problem of distinguisability of two states, $\hat{\rho}_0$ and $\hat{\rho}_1$, with their corresponding equiprobable binary hypothesis H_0 and H_1 respectively. However, so far, we have only discussed more of the formalism, but nothing that could be used quantify the distinction between two states. In particular, we want something that can quantify some *distance* between the two states or some kind of method to quantify some *error* for distinction between two states given the measurement. For the notion of *distance*, we use *fidelity of states* and as discused in chapter 1, the *Helstrom's Bound* can be viewed as a measure of the error which gives the minimum probability of error and the corresponding POVM that gives the minimum measurement errors.

3.3.1 Helstrom's Optimal Discrimination

As discussed in chapter 1, Helstrom's Bounds corresponds to the generic optimization problem to find the POVM that gives the best measurement with minimum error probability. Given the two states, we consider a dichotomous POVM $\{\hat{E}_0, \hat{E}_1\}$, where $\hat{E}_0$ corresponds to the hypothesis $\hat{H}_0$ and $\hat{E}_1$ corresponds to the hypothesis $\hat{E}_1$ respectively. The error probability can simply be obtained from the hypothesis testing for H_0 and H_1 as

$$P_E = \frac{1}{2}(P(H_0|H_1) + P(H_1|H_0)) \tag{3.41}$$

where,
$P(H_0|H_1)$ is the probability of concluding that H_0 is true, when H_1 is and
$P(H_1|H_0)$ is the probability of concluding that H_1 is true, when H_0 is true and
the factor of $\frac{1}{2}$ arises from the equiprobable assumptions.

Also, note that the expectation of an operator $\hat{O}$ given the system is in state $\hat{\rho}$ is given by $\left\langle \hat{O} \right\rangle = tr(r\hat{h}o\hat{O})$. So, if the system is in $\hat{\rho}_j$ and we make a measurement with the operator with $\hat{E}_i$, then the probability of error in the measurement is given by

$$P(H_i|H_j) = tr(\hat{\rho}_j \hat{E}_i) \tag{3.42}$$

Thus, using eq 3.42 in eq 3.41, we get

$$P_E = \frac{1}{2}(tr(\hat{\rho}_1 \hat{E}_0) + tr(\hat{\rho}_0 \hat{E}_1)) \tag{3.43}$$

Also, since we defined our POVM to be the set $\hat{E}_0, \hat{E}_1$ and POVM are complete, so $\hat{E}_0 + \hat{E}_1 = I \implies \hat{E}_0 = I - \hat{E}_1$. So, eq 3.43 will be,

$$\begin{aligned} P_E &= \frac{1}{2}(tr(\hat{\rho}_1(I - \hat{E}_1)) + tr(\hat{\rho}_0 \hat{E}_1)) \\ &= \frac{1}{2}(tr(\hat{\rho}_1) - tr(\hat{\rho}_1 \hat{E}_1) + tr(\hat{\rho}_0 \hat{E}_1)) \\ &= \frac{1}{2}(1 - tr(\hat{\rho}_1 \hat{E}_1) + tr(\hat{\rho}_0 \hat{E}_1)), \text{since } tr(\hat{\rho}_i) = 1 \\ &= \frac{1}{2}(1 - tr((\hat{\rho}_1 - \hat{\rho}_0)\hat{E}_1)) \end{aligned} \tag{3.44}$$

Note, all we aim to is find the correct POVM set so that the error probability is minimum. For this case, as $\hat{E}_0 + \hat{E}_1 = I$, so, we can express $\hat{E}_0$ in terms of $\hat{E}_1$. Thus, we have to minimize P_E over the measurement operator $\hat{E}_1$. Define $\hat{\gamma} = \hat{\rho}_1 - \hat{\rho}_0$. Clearly, $tr(\hat{\gamma}) = tr(\hat{\rho}_1) - tr(\hat{\rho}_0) = 1 - 1 = 0$. Since, both $\hat{\rho}_0$ and $\hat{\rho}_1$ are both positive and semi definite, so the eigenvalues of $\hat{\gamma}$ can be both positive or negative. Since both $\hat{\rho}_0$ and $\hat{\rho}_1$ are hermitian, so $\hat{\gamma}$ is also hermitian, hence it has the spectral decomposition,

$$\hat{\gamma} = \sum_k \mu_k \left|k\right\rangle \left\langle k\right| \tag{3.45}$$

where μ_k are the real eigenvalues. Now expanding $\hat{E}_1$ in the same basis

$$\hat{E}_1 = \sum_{k,l} e_{kl} \left|k\right\rangle \left\langle l\right| \tag{3.46}$$

Hence,

$$\begin{aligned} tr(\hat{E}_1\hat{\gamma}) &= tr\left(\sum_{k,l} e_{kl} \left|k\right\rangle \left\langle l\right| \cdot \sum_m \mu_m \left|m\right\rangle \left\langle m\right|\right) \\ &= \sum_{klm} e_{kl}\mu_m \left\langle l|m\right\rangle \left\langle m|k\right\rangle \\ &= \sum_{klm} e_{kl}\mu_m \delta_{l,m}\delta_{m,k} \\ &= \sum_k e_{kk}\mu_k \end{aligned} \tag{3.47}$$

Since POVMs are positive valued, so e_{kk} is positive. Now, μ_k can be positive or negative, but since we are optimizing e_{kk}, if μ_k is negative, it will lead to negative propagation of the errors, thus for such cases e_{kk} must be 0. Hence the net contribution will come only from the positive e_{kk} values. Thus the optimal $\hat{E}_1$ is one that projects onto positive part of $\hat{\gamma}$, $\hat{\gamma}_+$ (say) which projects to the subspace by the eigenstates of $\hat{\gamma}$ with positive eigenvalues. Thus, we have $tr(\hat{\gamma}\hat{E}_1) = tr(\hat{\gamma}_+) = \frac{1}{2}||\hat{\gamma}||_1$ where $||\hat{\gamma}||_1 = tr(|\hat{\gamma}|) = tr(\sqrt{\gamma^\dagger\gamma})$. Thus, using this, eq 3.44 can be written as

$$P_E^{min} = \frac{1}{2}(1 - ||\hat{\rho}_1 - \hat{\rho}_0||_1) \tag{3.48}$$

However, the calculation of the minimum probability of error using the above hypothesis is only for a single particle system with two states. But, the same analysis and calculations for an N particle system would be very difficult. This is because as the number of particles in the system increases the Hilbert Space grows exponentially over N as the N copy density matrix resides in the tensor product state of dimension d^N, where d is the dimension (degrees of freedom) for a single particle state, which may not simplify for larger N values. Moreover, opimization of such d^N matrix set (POVM) would also be analytically as well as numerically very difficult. However, one can calculate the *Quantum Chernoff Bounds*, which gives the asymptotic dependency of N copys of the system on the probability of error. Then the quantum hypothesis test becomes

$$\begin{aligned} H_0 &: \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0 \\ H_1 &: \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 \end{aligned} \tag{3.49}$$

Thus for this case, the minimum probability of error is given by

$$P_{E,N}^{min} = \frac{1}{2}(1 - ||\hat{\gamma}_N||_1) \tag{3.50}$$

where $\hat{\gamma}_N = \hat{\rho}_1 \otimes ... \otimes \hat{\rho}_1 - \hat{\rho}_0 \otimes ... \otimes \hat{\rho}_0$

3.3.2 Fidelity

Fidelity between two states could be used as a notion of measure of the *distance* between two states. For two states, $\hat{\rho}_1$ and $\hat{\rho}_2$, fidelity is defined as,

$$F(\hat{\rho}_1, \hat{\rho}_2) = \left(tr \left(\sqrt{\sqrt{\hat{\rho}_1} \hat{\rho}_2 \sqrt{\hat{\rho}_1}} \right) \right)^2 \tag{3.51}$$

The properties of the above fidelity equation to be used as a viable measure of distance between two states are:

- $\mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2) \leq 1$
- $\mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2) = \mathrm{F}(\hat{\rho}_2,\, \hat{\rho}_1)$
- If $\hat{\rho}_1$ is a pure state, $\hat{\rho}_1 = |\psi_1\rangle \langle\psi_1|$, then $\mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2) = \langle\psi_1|\hat{\rho}_2|\psi_1\rangle$
- $\mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2)$ is invariant under unitary transformations on the state space.
- $\mathrm{F}(\hat{\rho}_1 \otimes \hat{\sigma}_1,\, \hat{\rho}_2 \otimes \hat{\sigma}_2) = \mathrm{F}(\hat{\rho}_1,\, \hat{\rho}_2)\ \mathrm{F}(\hat{\sigma}_1,\, \hat{\sigma}_2)$

3.3.3 Quantum Chernoff Bounds

So far, we have discussed how the error is quantified and methods of quantifying the error. But, how is the actual minimization of the errror probability occurs. Chernoff solved this problem in the asymptotic regime and gave the asymptotic dependence on the probability of error if we have N copies of the system, i.e. a N trail hypothesis testing.[6] He showed that the probability of error decreases exponentially with the number of tests performed N, i.e. $P_E \approx exp(-\kappa N)$. The optimal exponent κ arising in the asymptotic limit is called Chernoff Bounds[6]. But why is Chernoff Bounds even considered... This may go back to 'One of the virtues of the Chernoff bound is that it yields a very natural distance measure between probability distributions; it is essentially the unique distance measure in the ubiquitous situation of independent and identically distributed (i.i.d.) random variables.'[6]

The quantum chernoff bounds serves as the upper limit the the minimum probability of error 3.50, i.e.

$$P_{N,E}^{min} \leq P_{QC}^{N} = \frac{1}{2} exp(-\kappa N) \tag{3.52}$$

where $\kappa = -ln \left(\min_{\,0 \leq s \leq 1} tr(\hat{\rho}_0^s \hat{\rho}_1^{(1-s)}) \right)$

Before diving deep into the quantum chernoff bounds, let us discuss a bit of the classical chernoff bounds. Consider a random variable X and $a \in \mathbb{R}$. Then, [7]

$$\begin{aligned} P(X \geq a) &\leq min_{s>0} e^{-sa} M_X(s) \\ P(X \leq a) &\leq min_{s<0} e^{-sa} M_X(s) \end{aligned} \tag{3.53}$$

For the optimal decision, we define the log likelihood,

$$L(x) = log\frac{p_1(x)}{p_0(x)}$$

For N trial observations

$$X = \sum_{i=1}^{N} L(x_i) = \sum_{i=1}^{N} log\frac{p_1(x_i)}{p_0(x_i)}$$

where we declare H_1 is true if $X > 0$ else we declare H_0 is true. Now we refer to the eq 3.41, which using above under hypothesis H_1

$$P(H_0|H_1) = P(X \leq 0|H_1) \leq e^{-s\cdot 0} M_X(s) = M_X(s)$$

where $s < 0$ The moment generating function $M_X(s)$ is defined as $\langle e^{sX} \rangle$, Hence

$$\begin{aligned} M_X(s) = \langle e^{sX} \rangle &= \langle e^{s\sum_{i=1}^{N} log\frac{p_1(x_i)}{p_0(x_i)}} \rangle \\ M_X(s) &= \langle e^{\sum_{i=1}^{N} log\frac{p_1(x_i)}{p_0(x_i)}^{s}} \rangle \\ M_X(s) &= \langle \prod_{i=1}^{N} e^{log\frac{p_1(x_i)}{p_0(x_i)}^{s}} \rangle \\ M_X(s) &= \prod_{i=1}^{N} \langle \frac{p_1(x_i)}{p_0(x_i)}^{s} \rangle \end{aligned} \tag{3.54}$$

Since, each factor is identical and $\langle f \rangle = \sum_{x_i} p(x)f(x)$ so,

$$M_X(s) = \left(\sum_x p_1(x) \left(\frac{p_1(x)}{p_0(x)} \right)^s \right)^N = \left(\sum_x p_1(x)^{1+s} p_0(x)^{-s} \right)^N \tag{3.55}$$

Taking $s \to -s$, i.e. $s > 0$

$$M_X(s) = \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{3.56}$$

$$P(H_0|H_1) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{3.57}$$

Similarly for $P(H_1|H_0)$, we get

$$P(H_1|H_0) \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{3.58}$$

where $s > 0$ Combining eq 3.57 and eq 3.58 with eq 3.41

$$P_E^N \leq \left(\sum_x p_1(x)^{1-s} p_0(x)^s \right)^N \tag{3.59}$$

Notice this takes the same form as the quantum Chernoff Bounds upto a factor of $\frac{1}{2}$, where instead of the probability desity functions $p_0(x)$ and $p_1(x)$, we have the density matrices ρ_0 and ρ_1.

One of the notable observation is that the Chernoff Bounds are exponentially tight. As the number of trails increases i.e. $N \to \infty$, $ln(P_{N,E}^{min})/N \to -\kappa$

Also, defining $Q_s = tr(\hat{\rho}_0^s \hat{\rho}_1^{1-s})$ and $Q = \min_{0 \leq s \leq 1} Q_s$, the quantum Chernoff Bounds can be written as

$$P_{QC}^N = \frac{1}{2} Q^N \tag{3.60}$$

The aim to find the minimum error would require the minimization of Q_s over all valures of $s \in [0, 1]$. Setting s = $\frac{1}{2}$, we will arrive at the *Quantum Bhattacharyya Bound*, however this bound is not assymptotically tight as the Quantum Chernoff Bound.

$$P_B^N = \frac{1}{2} \left(tr(\sqrt{\hat{\rho}_0 \hat{\rho}_1}) \right)^N \tag{3.61}$$

If the two states are indistinguisable, i.e. $\hat{\rho}_0 = \hat{\rho}_1$, then $P_{QC}^N \approx P_B^N$ and

$$\frac{1 - \sqrt{1 - F(\hat{\rho}_0, \hat{\rho}_1)}}{2} \leq P^{(1)} \leq P_{QC}^{(1)} \leq \frac{\sqrt{F(\hat{\rho}_0, \hat{\rho}_1)}}{2} \tag{3.62}$$

Also, it is noted that Q_s is a convex function over s and joint-concave over $\hat{\rho}_0$ and $]r\hat{h}o_1$. Q_s can be written as

$$Q_s = tr(\hat{\rho}_0^s \hat{\rho}_1^{1-s}) = \sum_k e_k^{(0)^s} e_k^{(1)^{1-s}}$$

where $e_k^{(0)}$ and $e_k^{(1)} \geq 0$ Thus each element of Q_s is $f_k(s) = a_k^s b_k^{1-s} = e^{slna_k} e^{(1-s)lnb_k}$, taking the 2nd derivative of f_k w.r.t. s, we see $f_k^{''}(s) = (lna_k - lnb_k)^s \cdot f_k(s) \geq 0$. Hence Q_s is convex over s. Also using Lieb's Theorem [8]

Theorem 3.2. *Lieb's Theorem: Let K be a fixed matrix in $\mathbb{C}^{n \times m}$. Then for any $t \in [0, 1]$, the map*

$$(A, B) \to tr[K^* A^{1-t} K B^t]$$

is jointly concave in (A, B) where A and B are respectively $n \times n$ and $m \times m$ Hermitian positive definite matrices.

Taking $A = \hat{\rho}_0$ and $B = \hat{\rho}_1$, and $K = \mathbb{I}$ and $t = s$, then Q_s is concave over $\hat{\rho}_0$ and $\hat{\rho}_1$.

3.4 Fidelity for Gaussian States

3.4.1 Single Mode Gaussian States

For two single mode gaussian thermal states with mean 0 and covariance matrices σ_1 and σ_2 corresponding to states $\hat{\rho}_1$ and $\hat{\rho}_2$ is given by

$$F_0(\hat{\rho}_1, \hat{\rho}_2) = \frac{2}{\sqrt{det(\sigma_1 + \sigma_2)}(\sqrt{det(\theta)} - \sqrt{det(\theta) - 1})} \tag{3.63}$$

where,
$det(\theta) = 1 + \frac{P}{det(\sigma_1+\sigma_2)}$
$P = (det(\sigma_1) - 1)(det(\sigma_2) - 1)$
It was also derived in the paper, the fidelity for a two single mode displaced squeezed thermal states. By the Williamson's Theorem, it could be seen that any arbitrary single mode gaussian state can be expressed as a displaced squeezed thermal state, i.e. $\hat{\rho}_G = \hat{U}_S \hat{\rho}_{th} \hat{U}_S^\dagger$. Thus, the fidelity for any single mode gaussian state,

$$F(\hat{\rho}_1, \hat{\rho}_2) = F_0(\hat{\rho}_1, \hat{\rho}_2) exp\left(-\frac{1}{2}(\overline{R_1} - \overline{R_2})^T \kappa (\overline{R_1} - \overline{R_2})\right) \tag{3.64}$$

where $F_0(\hat{\rho}_1, \hat{\rho}_2)$ is defined as eq 3.63 and $\kappa = \mathbb{I} - (\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$

3.4.2 Multimode Gaussian States

For comparing two multi-mode gaussian states, the fidelity is defined as [9]

$$F(\hat{\rho}_1, \hat{\rho}_2) = \sqrt{L det(\Phi(\mathcal{O}))}$$

where,
$\Phi(A) = A(\mathbb{I} + \sqrt{\mathbb{I} + (JA)^{-1}})$
$L^{-1} = det(\Phi(\sigma_1))^{-1} det(\frac{\Phi(\sigma_1)+\sigma_2}{2}) det(\frac{\sigma_2+\Phi(\sigma_1)-U}{2})$
$U = (\sigma_1 - i\Omega)(\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)$
$\mathcal{O} = \Phi(\sigma_1) - (\Phi(\sigma_1) - i\Omega)\left(\sigma_2 + \Phi(\sigma_1) - (\sigma_2 - i\Omega)(\Phi(\sigma_1) + \sigma_2)^{-1}(\sigma_2 + i\Omega)\right)^{-1}(\Phi(\sigma_1) + i\Omega)$

3.5 Quantum Chernoff Bounds for Gaussian States

Consider two n-mode Gaussian states, viz $\hat{\rho}_A$ and $\hat{\rho}_B$ with symplectic decomposition matrix S_A and S_B respectively, with symplectic eigenvalues $\{\alpha_k, k = 1, ..., n\}$ and $\{\beta_k, k = 1, ..., n\}$ respectively. The mean in the phase space are $\bar{x_A}$ and $\bar{x_B}$ respectively.

We know by Williamson's Theorem that every n-mode gaussian state ρ can be specified by a displacement vector $\bar{x} \in \mathcal{R}^{2n}$, a symplectic matrix S that preserves Ω and the symplectic eigenvalues $\nu_k, k = 1, ..., n$, where $\nu_k \geq 1$ and via the dicomposition

$$\rho = \hat{U}_{\bar{x}S}\rho_{th}\hat{U}^{\dagger}_{\bar{x}S}$$

where $\rho_{th} = \bigotimes_{k=1}^{n} \sigma\nu_k$ is the thermal state. Now,

$$\sigma(\nu_k) = \frac{2}{\nu_k + 1}\sum_{j=0}^{\infty}\left(\frac{\nu_k - 1}{\nu_k + 1}\right)^j |j\rangle\langle j|$$

. The covariance matrix is

$$V = S\left[\bigoplus_{k=1}^{n} \nu_k I_k\right] S^{\dagger}$$

[10][eq23, eq32]

Using *Lemma 1* of [10],

$$\rho^p = tr(\rho^p)\rho(p)$$

where $tr(\rho^p) = \prod_{k=1}^{n} G_p(\nu_k) = \prod_{k=1}^{n} \frac{2^p}{(\nu_k+1)^p - (\nu_k-1)^p}$. [10][eq30]

Then,

$$Q_s = tr(\hat{\rho}_A^s \hat{\rho}_B^{1-s}) = \prod_{k=1}^{n} G_s \alpha_k G_{1-s}(\beta_k) tr(\hat{\rho}_A(s)\hat{\rho}_B(1-s)) \tag{3.65}$$

with covariance matrices $V_A(s)$ and $V_B(1-s)$.

Also, from chapter 2 we have

$$tr(\hat{O}_1, \hat{O}_2) = \int_{\mathbb{C}^n} \frac{d^{2n}\vec{\lambda}}{\pi^n} \chi[\hat{O}_1](\vec{\lambda})\chi[\hat{O}_2](-\vec{\lambda})$$

Using the characteristics function for the gaussian state $\chi(\vec{\lambda}) = exp(-\frac{1}{2}\vec{\lambda}^T\sigma\vec{\lambda} + i\vec{\lambda}^T\bar{R})$ we get,

$$tr(\hat{\rho}_A, \hat{\rho}_B) = \int_{\mathbb{C}^n} \frac{d^{2n}\vec{\lambda}}{\pi^n} exp\left(-\frac{1}{2}\vec{\lambda}^T(V_A + V_B)\vec{\lambda} + i(\bar{x}_A - \bar{x}_B)\vec{\lambda}\right) \tag{3.66}$$

This is a standard integral. Solving we get,

$$tr(\hat{\rho}_A, \hat{\rho}_B) \propto exp\left(-\frac{1}{2}(\vec{x}_A - \vec{x}_B)^T(V_A + V_B)^{-1}(\vec{x}_A - \vec{x}_B)\right)$$

$$tr(\hat{\rho}_A, \hat{\rho}_B) \propto exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

where $\vec{d} = \vec{x}_A - \vec{x}_B$

Hence,

$$Q_s = tr(\hat{\rho}_A^s \hat{\rho}_B^{1-s}) \propto \prod_{k=1}^{n} G_s \alpha_k G_{1-s}(\beta_k) exp\left(-\frac{1}{2}\vec{d}^T(V_A + V_B)^{-1}\vec{d}\right)$$

$$Q_s = tr(\hat{\rho}_A^s \hat{\rho}_B^{1-s}) = \bar{Q}_s exp\left(-\frac{1}{2}\vec{d}^T (V_A + V_B)^{-1}\vec{d}\right)$$

It can further be calculated to find the value of $\bar{Q}_s$ as

$$\bar{Q}_s = \frac{2^n \prod_{k=1}^n G_s(\alpha_k) G_{1-s}(\beta_k)}{\sqrt{\det\left(V_A(s) + V_B(1-s)\right)}}$$

where,

$$V_A(s) = S_A \left(\bigotimes_{k=1}^n \Lambda_s(\alpha_k)\mathbb{I}_k\right) S_A^T$$

$$V_B(1-s) = S_B \left(\bigotimes_{k=1}^n \Lambda_{1-s}(\beta_k)\mathbb{I}_k\right) S_B^T$$

$$G_p(x) = \frac{2^p}{(x+1)^p - (x-1)^p}$$

$$\Lambda_p(x) = \frac{(x+1)^p + (x-1)^p}{(x+1)^p - (x-1)^p}$$

The above implementation can be used numerically to calculate the quantum Chernoff Bounds. However, the analytical solution, for most cases would be difficult because the minimization over s, which is required would be difficult for most physical systems due to the scale of the physical systems.

3.5.1 Numerical Analysis of Quantum Chernoff Bounds

Below is a numerical implimentation for the validation of the Helstrom Bounds over the limit of N. The limit of N taken here is to be N = 10.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize_scalar
from qutip import rand_dm, ket2dm, tensor
```

Listing 1: Includes

```
rho0 = rand_dm(2, seed=809)
rho1 = rand_dm(2, seed=593)
print(rho0)
print(rho1)
```

Listing 2: Defining the two state with random density matrix

Defining the two random density matrices

$$\hat{\rho}_0 = \begin{pmatrix} 0.5276 & -0.4914 + 0.0466i \\ -0.4914 - 0.0466i & 0.4724 \end{pmatrix}$$